KUDUM YASHWANTH

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PROFESSIONAL SUMMARY

Electronics and Communication Engineer with a strong foundation in Digital VLSI design, RISC-V SoC integration, ASIC physical design (RTL-to-GDSII), and Embedded AI systems. Designed and signed off Aurora v1 — a multi-clock RISC-V + AI accelerator System-on-Chip — through the complete RTL-to-GDSII flow on the SkyWater Sky130 PDK, achieving multi-corner timing closure (WNS/TNS = 0), clean DRC, and device-perfect LVS. Hands-on with SystemVerilog RTL, AMBA AXI4/AHB/APB protocols, UVM-style verification, and the Cadence (Genus/Innovus/Xcelium) and OpenLane toolchains. Also experienced in hardware–software co-design, AI inference acceleration (TensorRT, YOLOv8), ROS 2 robotics, and real-time embedded firmware development.

EDUCATION

B.E. in Electronics & Communication Engineering *Sri Sairam College of Engineering, VTU — Bengaluru* 2022 – 2026

Coursework: VLSI Design · Computer Architecture · Embedded Systems · Digital Signal Processing · AI Fundamentals

TECHNICAL SKILLS

Digital Design & Microarchitecture: SystemVerilog, Verilog, FSM Design (Moore/Mealy), Pipelining & Control Logic, Systolic Arrays, Sequential MAC Datapaths, Dataflow/Streaming Architectures, RISC-V SoC Integration

Protocols / SoC: AMBA AXI4, AXI4-Lite, AXI4-Stream, AHB, APB, Ready/Valid Handshake, Backpressure, Clock-Domain Crossing (CDC)

Verification: UVM-style Constrained-Random Testbenches, Functional Coverage, Assertions, Waveform Debugging, Protocol Compliance

ASIC & Physical Design: RTL-to-GDSII, Synthesis, Floorplan, Placement, CTS, Routing, Multi-Corner STA (Setup/Hold/Slack), DRC/LVS, PDN; OpenLane/OpenROAD (Sky130), Cadence Genus/Innovus/NCLaunch (Xcelium)/Virtuoso, Yosys, Magic, netgen, KLayout

AI & Acceleration: CNN Inference Pipelines, TensorRT Model Optimization, YOLOv8, DeepLabV3+, Edge-AI Deployment, ROS 2 Navigation & Sensor Fusion (EKF/SLAM)

Embedded Systems & Edge Computing: ESP32 Firmware (C/C++), Raspberry Pi (Linux), UART/SPI/I²C/I²S/PWM, Interrupt-driven Firmware, Hardware–Software Integration & Debugging

FPGA / Tools: Vivado, ModelSim, GOWIN IDE, Verilator, Docker, KiCad, Altium Designer, Flask, Git, VS Code, Linux

Programming Languages: Python, C, C++, Chisel

Languages: English, Kannada, Telugu, Hindi

INTERNSHIP

VLSI Design Intern *Maven Silicon Pvt. Ltd., Bengaluru* 04/2025 – 05/2025

- Designed and implemented an AHB-to-APB bridge compliant with the AMBA 4 protocol, with FSM-based state transitions for address, data, and handshake phases.
- Created a UVM-style constrained-random testbench with functional coverage and performed waveform-based debugging for handshake timing and protocol violations.
- Strengthened understanding of bus arbitration, latency handling, and SoC interconnect behavior.

PROJECTS

Aurora v1 — RISC-V + AI Accelerator SoC (RTL-to-GDSII) *SystemVerilog | AMBA AXI4 | Sky130 | OpenLane/OpenROAD* 2026

- Designed and integrated a multi-clock System-on-Chip pairing a Rocket RV64IMAC RISC-V CPU tile with a 4×4 systolic int16 matrix-multiply accelerator over an 8×8 AXI4 crossbar fabric (boot ROM, SRAM, UART, GPIO, timer, interrupt controller), and carried it through the complete RTL-to-GDSII flow.

- Achieved post-route, multi-corner static timing closure (WNS/TNS = 0 at slow/typical/fast corners), clean DRC, and device-perfect LVS (137,562 = 137,562 devices) on the full-chip Sky130 layout (41.25 mm^2 die, ~130k standard cells, 5-layer metal stack).
- Bridged a 33 MHz CPU domain and a 50 MHz fabric/accelerator domain with dual asynchronous FIFO clock-domain-crossing (CDC) bridges; hardened the CPU and accelerator as reusable AXI4 macros via a TileLink-to-AXI4 bridge.
- Resolved routing-congestion, power-grid/LVS, and timing-closure issues; verified full-SoC bring-up in simulation — boots, runs a matrix multiply on the accelerator, reads results back over AXI, zero traps.
- Closed the entire physical-design flow on commodity hardware using 100% open-source tools: Verilator, Yosys, OpenLane/OpenROAD, Magic, netgen, KLayout.

AI Accelerator for Neural Network Processing (ASIC) *SystemVerilog | AXI4-Stream | Cadence | OpenLane (Sky130)*

- Designed and implemented a modular CNN inference accelerator (image preprocessing, sequential MAC-based convolution engine, ReLU, pooling, fully connected layer) and completed a full RTL-to-GDSII ASIC flow.
- Built a 32-bit AXI4-Stream interface with ready/valid handshake and backpressure-aware pipeline stalling, using sequential MAC reuse for an area–performance tradeoff with parameterized RTL modules.
- Verified in Vivado and Cadence Xcelium — validated FSM transitions, AXI transactions, convolution correctness, and pipeline synchronization.
- Physical results (Sky130/OpenLane): 100 MHz target (1.08 ns critical path), WNS/TNS = 0, 4.0 mm^2 die / 3.23 mm^2 core, ~324k physical cells, 32,320 µm routed wirelength, 0 DRC violations, LVS clean.

Autonomous Exploration Rover — “The Thousand Sunny” *ROS 2 | Jetson Orin NX | SLAM | TensorRT*

- Designed an AI-driven 6WD rover integrating a Jetson Orin NX for on-board AI processing.
- Implemented a ROS 2 Humble navigation stack with SLAM (RTAB-Map, Cartographer) and multi-sensor fusion (IMU + LiDAR + RGBD + EKF).
- Deployed YOLOv8 and DeepLabV3+ optimized via TensorRT for real-time inference; built a hardware-in-loop test environment in NVIDIA Isaac Sim.
- Developed a telemetry dashboard using Flask with LoRa + Wi-Fi communication.

AI Personal Assistant *ESP32-based Embedded System*

- Designed a portable voice-controlled AI assistant integrating an ESP32 with an I^2S microphone and audio amplifier, with interrupt-driven firmware for button-triggered audio capture.
- Implemented buffered Wi-Fi API communication for AI response generation and a real-time audio pipeline ensuring a stable capture-process-playback cycle.
- Modified an open-source PCB design to optimize power delivery and signal integrity; added recovery logic and status indication for standalone stability.

AHB-to-APB Protocol Bridge *SystemVerilog | AMBA 4*

- Designed FSM-based protocol translation between a high-performance AHB bus and a low-speed APB bus, with wait-state handling and handshake timing compliance.
- Implemented correct sequencing for HADDR, HWRITE, HREADY, PSEL, PENABLE, and PREADY signals.
- Verified using a constrained-random testbench with assertion-style checks, achieving full functional coverage and zero handshake violations.

CERTIFICATIONS

- Samsung Chip Design Studio — IIIT Bangalore (2026)
- VLSI Design Internship — Maven Silicon (2025)
- NPTEL: System Design through Verilog (2025)
- NASA International Space Apps Challenge — Galactic Problem Solver (2025)